

LISTENING

#28

1 – Listen to a science interview about a Japanese astronaut who says he has grown taller in space. Pay attention to why astronauts get taller, how it affects their return to Earth, and what safety concerns are mentioned.

Growing in Space

1 - How long has Norishige Kanai been on the International Space Station?

2 - How much taller did he say he had become?

3 - Why is it difficult to measure height in space?

- a) There is no proper equipment
- b) There is no gravity to stand straight
- c) The tape measures are too old

4 - According to the former astronaut, what causes astronauts to become taller?

- a) Their bones grow longer
- b) Their backbone stretches without gravity
- c) They exercise more

5 - How many centimeters taller did the former astronaut say he became in space?

6 - Why is the seat size important when coming back to Earth?

- a) To make the flight more comfortable
- b) It affects the astronaut's safety during landing
- c) It is required by space law

7 - What body parts could be injured if the seat doesn't fit properly?

8 - How fast does the former astronaut say the body returns to normal height?

- a) Very slowly
- b) It depends on the person
- c) Quite rapidly

9 - What word does he use to describe gravity?

- a) A gentle force
- b) A punishing oppressor
- c) A helpful friend

Full Name _____

READING COMPREHENSION

What Happens to the Human Body in Space

Living in space isn't only a scientific achievement — it's also a challenge for the human body. Astronauts who stay in orbit for months experience a complete change in their environment. Without gravity, muscles and bones don't have to support the body's weight. As a result, they gradually weaken, which is why astronauts must exercise for hours every day on the International Space Station.

One surprising effect of microgravity is that astronauts become taller. When the force of gravity disappears, the spine stretches. The vertebrae, which are usually compressed on Earth, move slightly apart. In some cases, astronauts can gain several centimeters in height. However, this temporary height boost can lead to problems. Equipment such as specially designed seats for landing must be carefully fitted to avoid injury.

Space also affects the heart and blood flow. On Earth, gravity helps pull blood toward our legs. In space, more blood stays in the upper body, which can cause facial swelling and headaches. Over long periods, the heart doesn't work as hard to pump blood, so it becomes weaker.

Another major concern is radiation. Outside Earth's protective atmosphere, astronauts are exposed to higher levels of cosmic radiation. This increases the long-term risk of cancer.

Scientists are still studying these changes to make future space travel safer. Understanding how our bodies react to space will be essential if humans want to explore farther — perhaps even reaching Mars one day.

Retrieved from: <https://www.bbc.com/future/article/20180810-what-happens-to-your-body-in-spacea>

Comprehension Questions

1. Why do astronauts' muscles and bones become weaker in space?

2. What happens to the spine in microgravity?

3. What long-term health risk does space radiation increase?

4. Would you personally like to live in space for a long period? Why or why not?

5. What do you think would be the most difficult physical change to deal with in space?
